

Practical Week -2

Task-1

Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user Space and kernel space during execution.

```
GNU nano 8.7.1 task2.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd1, fd2;
    char buffer[100];
    int n;

    fd1 = open("File1.txt", O_RDONLY);

    if (fd1 < 0) {
        printf("Error opening File1.txt\n");
        return 1;
    }

    fd2 = open("File2.txt", O_CREAT | O_WRONLY | O_TRUNC, 0644);

    if (fd2 < 0) {
        printf("Error opening File2.txt\n");
        close(fd1);
        return 1;
    }

    n = read(fd1, buffer, sizeof(buffer));

    if (n < 0) {
        printf("Error reading file\n");
        close(fd1);
        close(fd2);
        return 1;
    }

    write(fd2, buffer, n);
    close(fd1);
    close(fd2);
    return 0;
}
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify

```

sirichandana@siri: ~  X  +

    if (n < 0) {
        printf("Error reading file\n");
        close(fd1);
        close(fd2);
        return 1;
    }

    write(fd2, buffer, n);

    close(fd1);
    close(fd2);

    printf("File copied successfully.\n");

    return 0;
}

^G Help      ^O Write Out  ^F Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File  ^\ Replace   ^U Paste     ^J Justify

sirichandana@siri:~$ echo "Hello, this is File1." >File1.txt
sirichandana@siri:~$ cat File1.txt
Hello, this is File1..
sirichandana@siri:~$ nano task2.c
sirichandana@siri:~$ gcc task2.c -o task2
sirichandana@siri:~$ ./task2
File copied successfully..
sirichandana@siri:~$ cat File2.txt
Hello, this is File1..
sirichandana@siri:~$ ls -l File.txt File2.txt
ls: cannot access 'File.txt': No such file or directory
File2.txt
sirichandana@siri:~$ ls -l File1.txt File2.txt
-rw-r--r-- 1 sirichandana sirichandana 22 Aug 16 11:27 File1.txt
-rw-r--r-- 1 sirichandana sirichandana 22 Aug 16 11:33 File2.txt

```

Task-2

Use the strace utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system 3 calls and identify the kernel services involved

```

sirichandana@siri: ~ | X +
sirichandana@siri:~$ echo "Hello Operating System" >sample.txt
sirichandana@siri:~$ cat sample.txt
Hello Operating System
sirichandana@siri:~$ strace -e trace=execve,openat,read,write,close,exit_group
at sample.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7ffc055e30c8 /* 25 vars */) = 0
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
close(3)
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0", 832) = 832
close(3)
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0", 832) = 832
close(3)
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\128\2\0\0\0\0\2\0", 832) = 832
close(3)
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\256\12\0\0\0\0\2\0", 832) = 832
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libcrt2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\6\0\0\0\0\0\0", 832) = 832
close(3)
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
read(3, "nodev\tsysfs\nodev\ttmpfs\nodev\tbd"..., 1024) = 442

```

```

sirichandana@siri: ~ | X +
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
read(3, "55c56b713000-55c56b805000 r--p 0" ..., 1024) = 1024
read(3, "0 08:30 13628" ..., 1024) = 1024
read(3, "00 08:30 13631" ..., 1024) = 1024
read(3, " /usr/lib/x86_64-lin" ..., 1024) = 1024
read(3, "0 13625 /us" ..., 1024) = 566
close(3) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/ucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
read(3, "# Common strings shared across a"..., 4080) = 4080
read(3, "", 32)
close(3)
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
openat(AT_FDCWD, "sample.txt", O_RDONLY|O_CLOEXEC) = 3
Hello Operating System
close(5) = 0
close(4) = 0
close(3) = 0
exit_group(0) ?
+++ exited with 0 +++
sirichandana@siri:~$

```